(1)

	September 2022					October 2022						November 2022				
M		5	12	19	26	31	3	10	17	24			7	14	21	28
T		6	13	20	27		4	11	18	25		1	8	15	22	29
W		7	14	21	28		5	12	19	26		2	9	16	23	30
T	1	8	15	22	29		6	13	20	27		3	10	17	24	
F	2	9	16	23	30		7	14	21	28		4	11	18	25	
S	3	10	17	24		1	8	15	22	29		5	12	19	26	
S	4	11	18	25		2	9	16	23	30		6	13	20	27	

23 Wednesday 327/038 — 23i-0683, Section-F — Labor Thanksgiving Day

8.00 am M.Ali Sher.

8.30

Question # 1

9.00

(a) considering instruction count as IC.

9.30

$$P1\ exec = \frac{IC \times 1.5}{2\times10^9} = I.C \times 0.75\ ns$$

10.00

$P2$ exec $= I.C \times 0.40\,ns$

10.30 $P3$ exec $= IC \times 0.88\,ns$

$P4$ exec $= IC \times 0.50\,ns$

11.00 $\therefore$ assume $IC = 1$ for now

P2 has least exec time so highest performance.

11.30 $$P2\ performance = \frac{1}{0.40\,ns} = \boxed{2.5\times10^9\ IPS}$$

Noon

(b) $$15 = \frac{IC \times 1.5}{2\times10^9} \quad \text{for } P1$$

12.30

$P1 \rightarrow 2\times10^{10}$ Instructions and $1.5\times2\times10^{10} = 3\times10^{10}$ cycle

1.00 similarly

1.30 $$P2 \rightarrow IC = \frac{15\times2.5\times10^9}{1} = 3.75\times10^{10}\ IC \text{ and } 3.75\times10^{10} \text{ cycles.}$$

2.00 $$P3 \rightarrow IC = \frac{15\times2.5\times10^9}{2.2} = 1.7\times10^{10} \text{ and } 3.75\times10^{10} \text{ cycles}$$

2.30 $$P4 \rightarrow IC = \frac{15\times4\times10^9}{2} = 3\times10^{10} \text{ and } 6\times10^{10} \text{ cycles.}$$

3.00 (c) exec time new $= 40\%$ exec time old.

Exec time depends on CPI, IC and clock Rate,

3.30 we can reduce CPI, IC and increase clock Rate

to achieve speedup.

4.00

$$\frac{1}{(1-0.4)} = \frac{1}{0.6} = 1.67\times \text{ increase in clock rate.}$$

4.30

5.00

6.00 pm

Question 2

a) M1 : $\cancel{50\%}$ $0.5\times6+0.3\times5+0.2\times2$
$= 4.9$ av CPI

M2 : $0.5\times1+0.3\times3+0.2\times4 = 2.2$ av CPI

b) MIPS (M1) $= \frac{1500 \text{ MHz}}{4.9\times10^6} = 306.1$ millions of inst. per sec (mips)

MIPS (M2) $= \frac{2000 \cancel{\text{M}}\text{Hz}}{2.2\times\cancel{10^6}} = 909.1$ mips

c) M1 has a smaller rating.
Since set A has highest freq, i'll tinker w/ that.
target ≥ 909.1 mips.

$909.1 < \frac{1500 \cancel{\text{M}}\text{Hz}}{(0.5x+1.9)\times\cancel{10^6}} \Rightarrow$

$454.5x + 1727.27 \leq 1500$
$\boxed{x \leq -0.5}$

which is not possible. Thus M1 cannot be improved upto or more than M2.

Question #3

a) A : $CPI_{av} = \frac{1.8}{3.2\times10^9\times0.5\times10^{-9}} = 1.125$ av CPI

B : $CPI_{av} = \frac{2.5}{2.1\times0.5} = 2.38$ av CPI

(b) P1 (A) : av CPI = 1.125 , IC : 3.2×10^9

P2 (B) : av CPI = 2.38 , IC : 2.1×10^9

exec 1 = exec 2

$$\frac{IC_1 \times CPI_1}{\text{Clock Rate}_1} = \frac{IC_2 \times CPI_2}{\text{Clock Rate}_2}$$

$$\frac{3.6}{CR_1} = \frac{4.998}{CR_2}$$

$$\frac{CR_2}{CR_1} = \frac{4.998}{3.6} = 1.38\times$$

$\boxed{\text{B is 1.38x faster than A}}$

(c) Since we concluded B is faster.

~~considering same execution times~~.

~~$1.5\times10^8 \times 1.1 = 4.998\times10^9$~~

$$exec_{new} = 1.5\times10^8 \times 1.1 \times 0.5\times10^{-9}\ s$$
$$= 0.0825\ s$$

$$exec_B = 2.1\times10^9 \times 2.38 \times 0.5\times10^{-9}\ s$$
$$= 2.5\ s$$

$$\frac{execB}{exec\,C} = 30.3\times \text{ faster than B}$$

$$\frac{exec_A}{exec_C} = \frac{3.2\times10^9 \times 1.125 \times 0.5\times10^{-9}}{0.0825} = 21.8\times$$

(4)

Question # 4

(a)

instruction	CPI	IC	2 GHz
arth A	1	2.56×10^9	$= 2\times10^9\ s^{-1}$
ld/st B	12	1.28×10^9	
branch C	5	2.56×10^8	

$0.7\times p$ IC

i) $P=1$: $\left(\frac{2.56}{2\times0.7} + \frac{12\times1.28}{2\times0.7} + \frac{5\times2.56}{20}\right) = 13.44\ s$

ii) $P=2$: $\left(\frac{2.56}{2\times0.7\times2} + \frac{12\times1.28}{0.7\times2\times2} + \frac{5\times2.56}{20}\right) = 7.04\ s$

iii) $P=4$: $\left(\frac{2.56}{2\times0.7\times4} + \frac{12\times1.28}{0.7\times2\times4} + \frac{5\times2.56}{20}\right) = 3.84\ s$

iv) $P=8$: $2.24\ s$ similarly.

$\text{Speedup}_{P=2} = 13.44/7.04 = 1.9x$

$\text{Speedup}_{P=4} = 13.44/3.84 = 3.5x$

$\text{Speedup}_{P=8} = 13.44/2.24 = 6x$

(b) $P=1$: $\left(\frac{2.56\times2}{2\times0.7} + \frac{12\times1.28}{2\times0.7} + \frac{5\times2.56}{20}\right) = 15.26\ s$

$P=2$: $\left(\frac{1}{2}\ '' + \frac{1}{2}\ '' + ''\right) = 7.95\ s$

$P=4$: $\left(\frac{1}{4}\ '' + \frac{1}{4}\ '' + ''\right) = 4.29\ s$

$P=8$: $\left(\frac{1}{8}\ '' + \frac{1}{8}\ '' + ''\right) = 2.46\ s$

(c) For P=4 time was 3.84 s

so

$$3.84 = \frac{2.56}{2\times0.7} + \frac{x\times1.28}{2\times0.7} + \frac{5\times2.56}{20}$$

$$\frac{1.071\times2\times0.7}{1.28} = x, \quad x = 1.5$$

The CPI of L/S ins must be 1.5 or lower.

Question # 5

IC : 2.389×10^{12}, time = 750s, ref time = 9650 s

(a) cycle time = 0.333 ns

$$CPI = \frac{750}{2.389\times10^{12}\times0.333\times10^{-9}} \approx 0.942$$

(b) $IC' = IC\times0.1 + IC$, $time' = IC'\times CPI\times$ cycle time

≈ 825.68 s

(c) IC′ as above and $CPI' = CPI\times0.05 + CPI$

time′ = IC′ × CPI′ × cycle time ≈ 866 s

(d) 4×10^{9} Hz

$IC = (2.389 - 2.389\times0.15)\times10^{12} = 2.03065\times10^{12}$

time = 700 s

$$CPI = \frac{700\times4\times10^{9}}{2.03065\times10^{12}} \approx 1.378$$

(e) CPI increase causes decrease in execution time, increase in clock rate has the opposite effect. Thus they are disimilar.

(f) $\frac{750}{700} \approx 1.07\times$ by 1.07 times (reduced) or

$$\frac{750-700}{750} \approx 6.67\%\text{ reduction}$$

August 2022					
M	1	8	15	22	29
T	2	9	16	23	30
W	3	10	17	24	31
T	4	11	18	25	
F	5	12	19	26	
S	6	13	20	27	
S	7	14	21	28	

September 2022					
M		5	12	19	26
T		6	13	20	27
W		7	14	21	28
T	1	8	15	22	29
F	2	9	16	23	30
S	3	10	17	24	
S	4	11	18	25	

October 2022						
M	31	3	10	17	24	
T		4	11	18	25	
W		5	12	19	26	
T		6	13	20	27	
F		7	14	21	28	
S	1	8	15	22	29	
S	2	9	16	23	30	

18 Tuesday 291/074

8.00 am

(g) 960 ns time, 1.61 CPI and 4 GHz

8.30

time' = [960 × (1 − 0.1)] ns = 864 ns

9.00

$$IC = \frac{864 \times 10^{-9}}{1.61} \times 4 \times 10^9 = 2146.5$$

9.30

10.00

(h) time' = 864 × (1 − 0.1) ns = 777.6 ns

10.30

$$\frac{2146.5 \times 1.61}{777.6 \times 10^{-9}} = 4.44 \times 10^9\ s^{-1}$$

11.00

$$= 4.44\ GHz$$

11.30

(i) CPI' = 1.61 × (1 − 0.15) = 1.3685

time' = 960 × (1 − 0.2) ns = 768 ns

Noon

IC = 2146.5

12.30

$$\frac{2146.5 \times 1.3685}{768 \times 10^{-9}} = 3.82\ GHz$$

1.00

Question # 6

1.30

(a) P1 : performance = $\dfrac{4 \times 10^9}{5 \times 10^9 \times 0.9} = 0.88\ s^{-1}$

2.00

2.30

P2 : perf. = $\dfrac{3 \times 10^9}{10^9 \times 0.75} = 4\ s^{-1}$

3.00

Performance P1 < Prerformance P2 although
3.30
Clock Rate P1 > Clock Rate P2, so
it does not hold, it depends on IC and CPI as well.

4.00

4.30

(b) P1 time = $\dfrac{0.9}{4} = 0.225 = \dfrac{IC \times 0.75}{3 \times 10^9}$

5.00

$$\frac{0.225 \times 3 \times 10^9}{0.75} = IC \Rightarrow 9 \times 10^8\ ICs$$

6.00 pm

(c) P1 : $MIPS = \frac{5\times10^9}{1.125s\times10^6} = 4444.4$ MIPS

P2 : $MIPS = \frac{10^9}{0.25\times10^6} = 4000$ MIPS

False again as MIPS P1 > MIPS P2
but Perf. P1 < Perf. P2

(d) 40% of ICs are MFLOPs in both.

P1 : $\frac{5\times10^9 \times 0.4}{1.125\times10^6} = 1777.8$ MFLOPS

P2 : $\frac{10^9 \times 0.4}{0.25\times10^6} = 1600$ MFLOPS

Same as in part (c).

Question #7

(a) $90(1-0.2) + 120 + 70 = 262$ s.

$\frac{280-262}{280} = 6.4\%$ reduce in total time

(b) Total time new $= 280(1-0.2) = 224$ s
effect on Float op. (There are no int operations) = ?
assuming all others remain same.

$90(x) + 120 + 70 = 224$
$x = \frac{34}{90} = 0.37777$
$x = 0.37777 = (1-0.62222)$
62% reduction in FP time

(c) 224 s : $90 + 120 + 70x$
$\frac{14}{70} = x$, $x = 0.2 = 1-0.8$

8

20 Thursday 293/072

8.00 am an 80% reduce is required

8.30 Question # 8

Ins type	IC	CPI	2.5 GHz
FP	4800	1	
int	6000	1	
L/S	10800	4	
branch	1800	2	

(a) $time = \frac{(4800 + 6000 + 10800 \times 4 + 1800 \times 2)}{2.5 \times 10^9}$

$= 23040 \text{ ns}$

$time' = \frac{23040 \text{ ns}}{2} = 11520 \text{ ns}$

$11520 \times 10^{-9} = \frac{4800x}{2.5 \times 10^9} + 21120 \times 10^{-9}$

$(2.5)(-9600) = 4800x$, $x = -5$

CPI of FP must be -5, which is NOT possible.

(b) $11520 \times 10^{-9} = 5760 \times 10^{-9} + \frac{10800x}{2.5 \times 10^9}$

$14400 = 10800x$

$x = 1.33$

CPI must be reduced to 1.33

(c) $time' = \frac{(4800 + 6000)(1 - 0.4) + (10800 \times 4 + 1800 \times 2)(1 - 0.3)}{2.5 \times 10^9}$

$= 1.569 \times 10^{-5} \text{ s}$

Question# 9

a) $\text{Speedup}(\text{div})_{\text{overall}} = \dfrac{1}{1-0.3 + \dfrac{0.3}{3}} = 1.25$ (division)

$\text{Speedup}(\text{mul})_{\text{overall}} = \dfrac{1}{1-0.45 + \dfrac{0.45}{5}} = 1.56$ (multiplication)

- Neither works.

b) Combine.

$\text{Speedup}_{\text{overall}} = \dfrac{1}{\dfrac{0.3}{3} + \dfrac{0.45}{5} + \dfrac{0.25}{1}} = 2.27x$

c) x part has 3x $\text{speedup}_{\text{enhanced}}$, $(1-x)$ part is other.

$\text{Speedup}_{\text{overall}} = 4 = \dfrac{1}{\dfrac{x}{3} + \dfrac{1-x}{1}} \Rightarrow \dfrac{1}{4} = \dfrac{x}{3} + 1 - x$

~~$4+1 = x - 4x$, $-3 = -3x$, $x = 1$~~

$0.75 = x + 3 - 3x \Rightarrow -2.25 = -2x$

$x = 1.125$

or 112.5% of code must be of FP

NOT possible.

10

	August 2022					September 2022						October 2022				
M	1	8	15	22	29	M		5	12	19	26	M	31	3	10	17
T	2	9	16	23	30	T		6	13	20	27	T		4	11	18
W	3	10	17	24	31	W		7	14	21	28	W		5	12	19
T	4	11	18	25		T	1	8	15	22	29	T		6	13	20
F	5	12	19	26		F	2	9	16	23	30	F		7	14	21
S	6	13	20	27		S	3	10	17	24		S	1	8	15	22
S	7	14	21	28		S	4	11	18	25		S	2	9	16	23

12 Wednesday 285/080

8.00 am Question # 10

8.30 part A

ⓐ x part is 5x (enhanced)

9.00 so

$$2 = \frac{1}{\frac{x}{5} + \frac{1-x}{1}} \Rightarrow \frac{1}{2} = \frac{x+5-5x}{5}$$

9.30

10.00 $-2.5 = -4x \quad , x = 0.625$

10.30 62.5% of code must use MMX

11.00 ⓑ After 2x speedup is achieved:

say og time is 1s.

11.30 $\rightarrow$ new time $= \frac{0.625}{5} + (1-0.625) = 0.5s$

Noon so $\frac{0.625}{5} \times \frac{1}{0.5} = 25\%$

12.30

ⓒ $\text{Speedup}_{\text{overall}} = \frac{1}{1/5} = 5\times$ for 100% MMx

1.00

1.30 for 2.5x, $2.5 = \frac{1}{\frac{x}{5} + \frac{5-5x}{5}}$, $2 = -4x+5$

2.00

$0.75 = x$ so 75% of code must be MMX.

2.30

part B

3.00 ICs are same. (IC=1)

A: time $= 25\times10^{-9} \times 2.5$ s $= 62.5$ns

3.30 B: time $= 10\times10^{-9} \times 1.5$ s $=$ 15 ns

4.00 $\frac{62.5-15}{62.5} = 76\%$, B is 76% faster than A

4.30 or B is 4.16x faster

5.00 This disproves that "A will always run faster than B".
B is faster regardless MIPS or clockrate because of
6.00 pm lower clock cycle time.